

PE-EC703C — Wavelet Transforms

Semester VII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Orthonormal MRA & Filter Banks

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	2-Channel Filter Banks and Subband Coding	2
1.1	Analysis and Synthesis Filter Banks	2
2	Perfect Reconstruction (PR) Conditions	2
2.1	Aliasing Cancellation Condition	2
2.2	No Distortion Condition	3
2.3	Orthonormal Quadrature Mirror Filters (QMF)	3
3	Discrete-Time Multi-resolution Analysis	3
4	Orthonormal Wavelet Families	3

1. 2-Channel Filter Banks and Subband Coding

Multi-resolution analysis is implemented in practice using **digital filter banks**. A **2-Channel Filter Bank** splits a signal into two subbands (Analysis) and reconstructs it back to a single signal (Synthesis).

1.1 Analysis and Synthesis Filter Banks

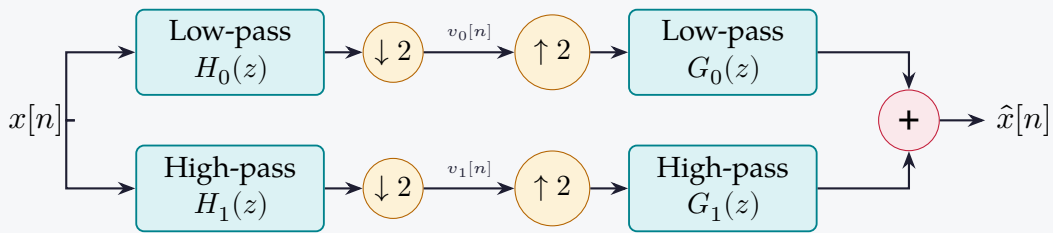
1. Analysis Filter Bank (Decomposition):

- **Low-Pass Filter ($H_0(z)$):** Extracts approximation components.
- **High-Pass Filter ($H_1(z)$):** Extracts detail components.
- **Decimation / Downsampling ($\downarrow 2$):** Discards every second sample to maintain the original data rate, yielding subband signals $v_0[n]$ and $v_1[n]$.

2. Synthesis Filter Bank (Reconstruction):

- **Interpolation / Upsampling ($\uparrow 2$):** Inserts a zero between adjacent samples.
- **Low-Pass Filter ($G_0(z)$):** Smooths the upsampled approximation.
- **High-Pass Filter ($G_1(z)$):** Smooths the upsampled detail.
- The filtered outputs are summed together to produce the reconstructed signal $\hat{x}[n]$.

2-Channel Subband Coder / Filter Bank System



2. Perfect Reconstruction (PR) Conditions

Let the Z -transform of the reconstructed signal be $\hat{X}(z)$. The input-output relationship of the 2-channel filter bank is:

$$\hat{X}(z) = \frac{1}{2} [H_0(z)G_0(z) + H_1(z)G_1(z)] X(z) + \frac{1}{2} [H_0(-z)G_0(z) + H_1(-z)G_1(z)] X(-z)$$

where the term involving $X(-z)$ represents **aliasing** introduced by downsampling.

2.1 Aliasing Cancellation Condition

To completely cancel aliasing, the second term must be zero for all $X(z)$:

$$H_0(-z)G_0(z) + H_1(-z)G_1(z) = 0$$

This is achieved by choosing synthesis filters that mirror the analysis filters:

$$G_0(z) = H_1(-z), \quad G_1(z) = -H_0(-z)$$

2.2 No Distortion Condition

After substituting the aliasing cancellation parameters, we require $\hat{x}[n]$ to be a delayed version of the input: $\hat{x}[n] = x[n-d] \implies \hat{X}(z) = z^{-d}X(z)$. This yields the **no distortion condition**:

$$H_0(z)G_0(z) + H_1(z)G_1(z) = 2z^{-d}$$

2.3 Orthonormal Quadrature Mirror Filters (QMF)

In an orthonormal filter bank, the filters must satisfy conjugate symmetry conditions. In the time domain, the relationship is:

$$h_1[k] = (-1)^k h_0[N-1-k] \quad (\text{where } N \text{ is the filter length})$$

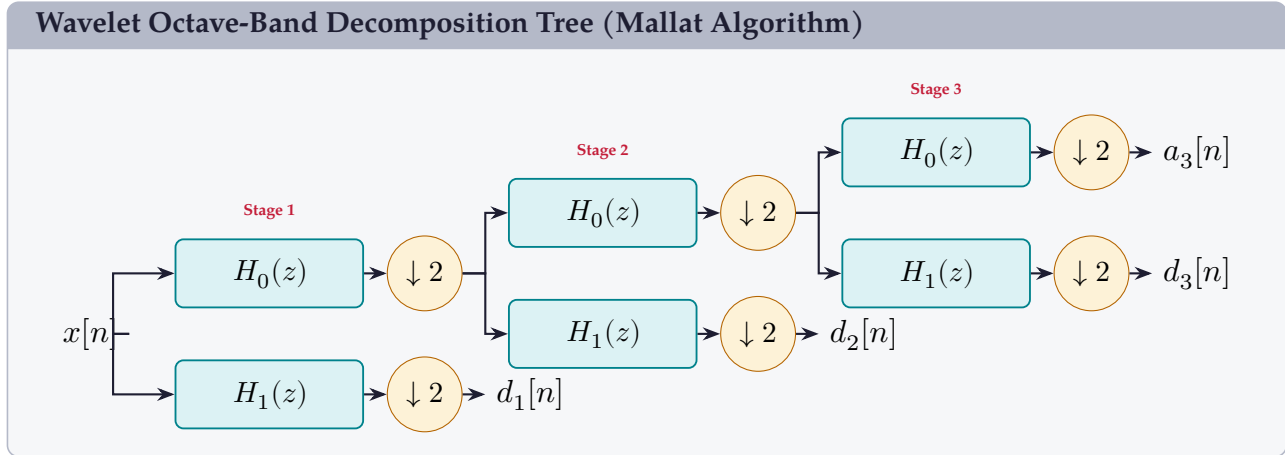
In the Z -domain, this translates to:

$$H_1(z) = z^{-(N-1)} H_0(-z^{-1})$$

This ensures that the filter bank cancels aliasing, has no amplitude distortion, and is orthonormal.

3. Discrete-Time Multi-resolution Analysis

To decompose a signal across multiple scales, we cascade the Analysis filter bank in an octave tree structure (the **Mallat Algorithm**).



4. Orthonormal Wavelet Families

To compute wavelets from filter coefficients, we use the **Cascade Algorithm (Refinement)**. By repeatedly upsampling and filtering $h_0[k]$, the scaling function $\phi(t)$ is generated at the limit.

Several standard orthogonal families exist, each offering unique trade-offs:

1. **Daubechies Wavelets (dbN)**: Orthonormal, compactly supported wavelets with the maximum number of vanishing moments (equal to N , the filter order). This maximizes frequency response sharpness and makes them excellent for compression.

2. **Symlets (symN):** Designed by Ingrid Daubechies to be "least asymmetric" versions of the db wavelets, providing near-linear phase characteristics while maintaining orthonormality.
3. **Coiflets (coifN):** Scaling functions also have vanishing moments, which improves polynomial approximation accuracy in both scaling and wavelet spaces.

Quick Revision — Module 3

Orthonormal MRA & Filter Banks | PE-EC703C

Key Filter Bank Equations

Aliasing Cancellation	$H_0(-z)G_0(z) + H_1(-z)G_1(z) = 0$
No Distortion Condition	$H_0(z)G_0(z) + H_1(z)G_1(z) = 2z^{-d}$
Synthesis Filters (PR)	$G_0(z) = H_1(-z), \quad G_1(z) = -H_0(-z)$
Orthonormal QMF Relation	$h_1[k] = (-1)^k h_0[N - 1 - k] \implies H_1(z) = z^{-(N-1)} H_0(-z^{-1})$
Vanishing Moments dbN	$\int_{-\infty}^{\infty} t^p \psi(t) dt = 0 \quad \text{for } p = 0, 1, \dots, N - 1$

One-Line Definitions

- **Filter Bank:** A system of parallel digital filters that splits (Analysis) and recombines (Synthesis) a signal.
- **Subband Coding:** Compression technique where a signal is split into frequency bands and coded separately.
- **Decimation ($\downarrow 2$):** Downsampling operation that discards every other sample, compressing the spectrum.
- **Interpolation ($\uparrow 2$):** Upsampling operation that inserts zeros between samples, expanding the spectrum.
- **QMF (Quadrature Mirror Filter):** A filter bank where high-pass and low-pass filters are mirror images across $f_s/4$.
- **Vanishing Moments:** The power to which a wavelet can cancel out polynomial components, aiding signal compression.
- **Cascade Algorithm:** Iterative upsampling/filtering technique used to construct scaling functions from coefficients.

Frequently Asked / Viva Questions

Q1. What are the conditions for Perfect Reconstruction (PR) in a 2-channel filter bank?

Perfect reconstruction requires: (1) Aliasing Cancellation: $H_0(-z)G_0(z) + H_1(-z)G_1(z) = 0$, and (2) No Distortion: $H_0(z)G_0(z) + H_1(z)G_1(z) = 2z^{-d}$, where d is the system delay.

Q2. How does downsampling introduce aliasing? How is it resolved in filter banks?

Downsampling by 2 discards alternate samples, which stretches the spectrum. If

Frequently Asked / Viva Questions (contd.)

the signal bandwidth exceeds $f_s/4$, spectral overlap (aliasing) occurs. In filter banks, the synthesis filters $G_0(z)$ and $G_1(z)$ are designed such that the aliased components from the low-pass and high-pass branches cancel each other out during summation.

Q3. Define the Quadrature Mirror Filter (QMF) relationship.

A QMF bank uses mirror-symmetric filters. The high-pass filter is a modulated version of the low-pass filter: $h_1[k] = (-1)^k h_0[N-1-k]$. In the frequency domain, $|H_1(e^{j\omega})| = |H_0(e^{j(\pi-\omega)})|$.

Q4. What are vanishing moments, and why are they important in Daubechies wavelets?

A wavelet has N vanishing moments if $\int_{-\infty}^{\infty} t^p \psi(t) dt = 0$ for $p = 0, \dots, N-1$. It means the wavelet ignores polynomials up to degree $N-1$. In Daubechies wavelets, having more vanishing moments allows the wavelet to represent smooth signals with fewer non-zero coefficients, maximizing compression.

Q5. What is the Mallat Algorithm?

The Mallat Algorithm (or Fast Wavelet Transform) is a recursive algorithm that computes DWT coefficients by cascading 2-channel filter banks. It splits the low-pass approximation at each stage into a new approximation and detail, achieving $O(N)$ computational complexity.

Q6. Why do we prefer Symlets over standard Daubechies wavelets in some applications?

Standard Daubechies wavelets are highly asymmetric, causing non-linear phase response and phase distortion in filtered signals. Symlets are designed to be "least asymmetric" while retaining compact support and orthonormality, providing near-linear phase.

Summary Comparison of Wavelet Families

Family	Symmetry	Compact Support	Vanishing Moments
Haar	Symmetric (linear phase).	Yes (shortest).	1 (poor smoothness).
Daubechies (dbN)	Asymmetric (non-linear phase).	Yes (width $2N-1$).	N (excellent compression).
Symlets (symN)	Near-symmetric (near-linear).	Yes (width $2N-1$).	N (linear phase trade-off).
Coiflets (coifN)	Near-symmetric (near-linear).	Yes (width $6N-1$).	$2N$ for wavelet, $2N-1$ for scaling.